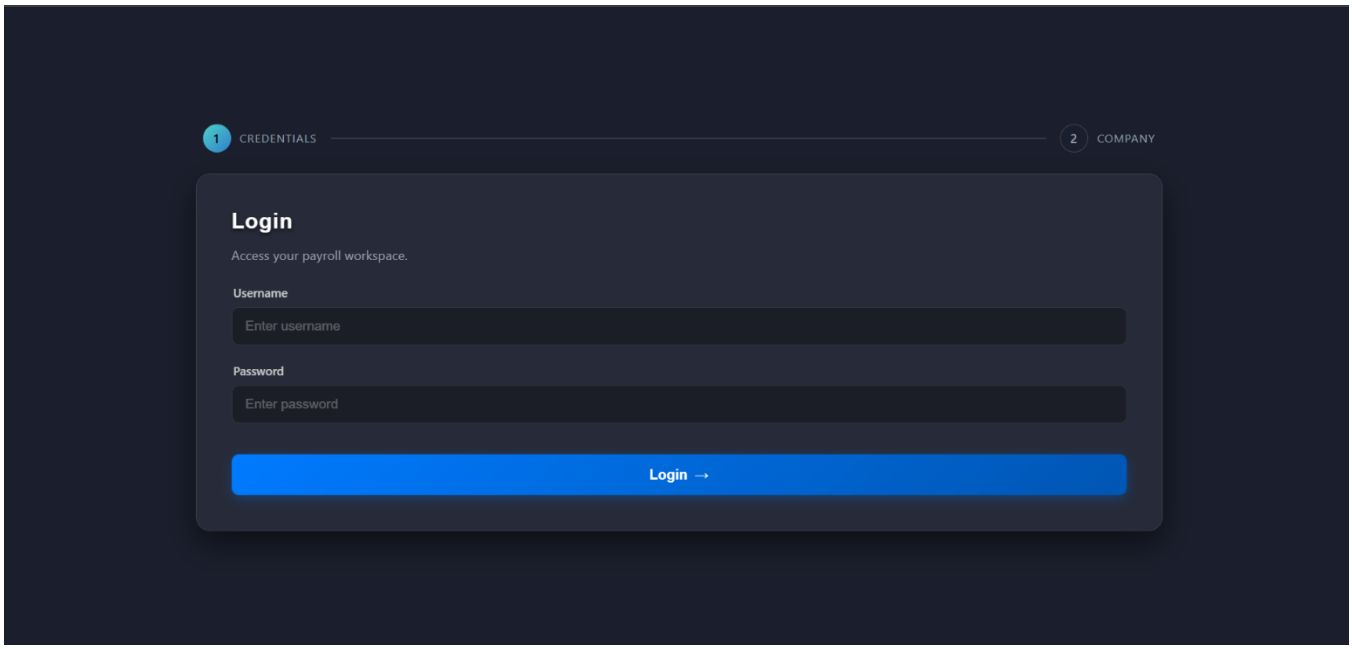
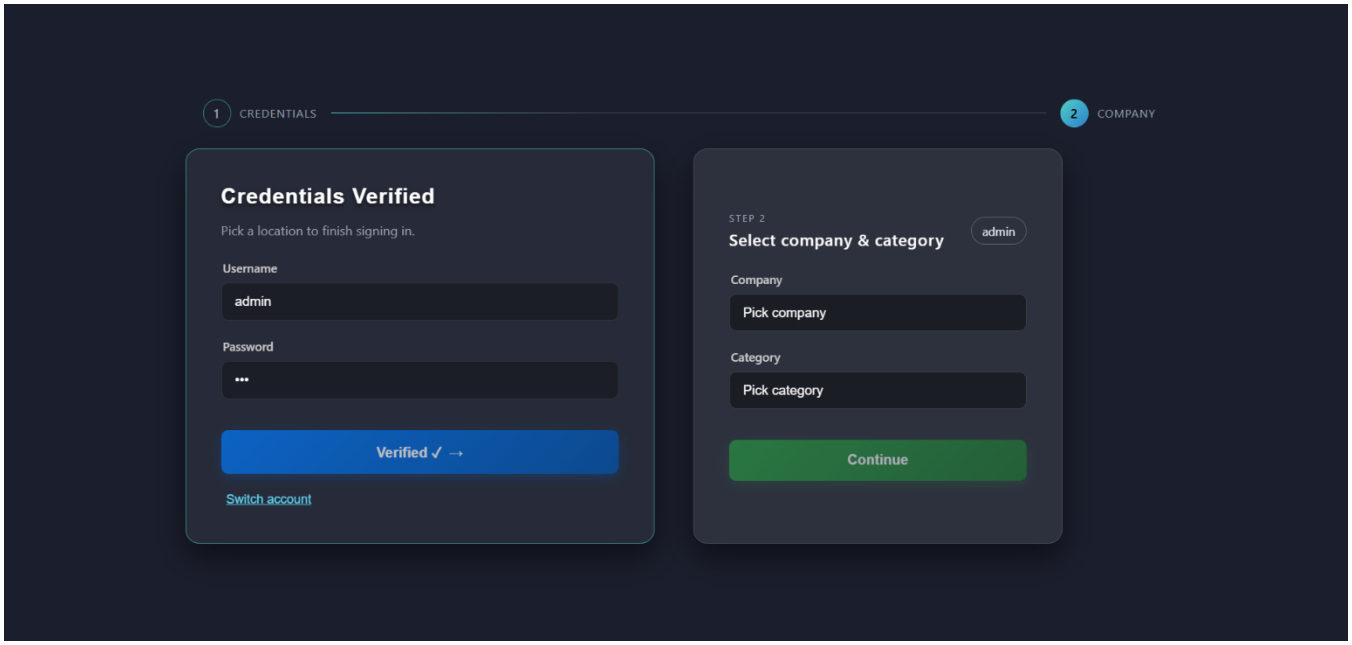
Web-Based Payroll System

VERSION: 1.00

System Overview

The Web-Based Payroll System is a secure, responsive single-page application featuring an intuitive sidebar navigation and modal popups, allowing HR and administrative staff to use the system easily and effectively. The series of activities are categorized and arranged in an orderly, logical flow beginning with Master Files (for Company, Employee, and Bank configurations), followed by daily Transactions (such as loans and salary changes), proceeding to core Processes (for automated month-end calculations including EPF/ETF and dynamic PAYE), and concluding with Reports (digital payslips and bank transfer lists) respectively. This structured layout enables end-users to transition smoothly from legacy workflows and get to the desired screen easily and more quickly.





Key System Features & Benefits

Automated Payroll Calculations - Efficiency and accuracy in payroll processing are critical. By automating the monthly calculation cycle, the system eliminates the need for manual, error-prone spreadsheet work. The platform seamlessly processes complex financial transactions, including basic pay adjustments, overtime, attendance allowances, and loan installments, saving the HR team significant time and administrative effort.

Dynamic Tax & Statutory Compliance - Computations for taxes and government statutory deductions can be complex and stressful. This system features automated tax management modules that minimize the burden of employee reporting obligations. All tax data and statutory deductions are calculated automatically and kept securely in the system, making it easy to pull accurate records for compliance without digging through piles of physical pay-slips.

Reliable Backup & Centralized Data Storing large amounts of sensitive payroll data manually or in fragmented files is risky. The application utilizes a centralized database to securely store all employee and financial records. With automated database backups, it is convenient to save records and ensures that, in the event of local hardware failure, all historical payroll data is preserved and fully recoverable.

Comprehensive & Instant Reporting - Generating required month-end documentation is reduced to a few clicks. The system automatically produces formatted digital payslips, customized bank transfer files, and mandatory government EPF/ETF reports. Historical data is securely archived upon month-end finalization, allowing HR to instantly pull past transaction records to quickly resolve any employee disputes.

Web-Based Accessibility & Usability - Unlike standalone desktop applications restricted to specific office computers, this modern web architecture allows authorized staff to securely manage payroll operations from anywhere using a standard web browser. The clean, responsive interface eliminates the steep learning curve of legacy software, making global employee updates and mass data management effortless.

Recommended Hardware & Software Requirements

Because this is a web-based application utilizing a Client-Server architecture, the system requirements are divided between the server environment (where the application and database are hosted) and the client environment (the end-users accessing the system).

Server Environment (Hosting)

Hardware Requirements:

- **Processor (CPU):** Quad-core processor (e.g., Intel Xeon or equivalent) @ 2.5 GHz or higher.
- **Memory (RAM):** 8 GB minimum (16 GB or higher recommended for optimal database performance during heavy month-end processing).
- **Storage:** 100 GB SSD (Solid State Drive) minimum for the operating system, application files, and database storage. SSD is highly recommended to ensure fast read/write speeds for database queries.
- **Network:** High-speed, stable internet connection with a static IP address.

Software Requirements:

- **Operating System:** Linux (e.g., Ubuntu Server 20.04 LTS or newer) or Windows Server (2019 or newer).
- **Web Server / Runtime:** Node.js environment (LTS version).
- **Database Management System:** MySQL (Version 8.0 or higher).
- **Security:** SSL/TLS Certificate for HTTPS encryption.

Client Environment (End-User)

Hardware Requirements:

- **Processor (CPU):** Dual-core processor (Intel Core i3 or equivalent).
- **Memory (RAM):** 4 GB minimum (8 GB recommended for smoother multitasking).
- **Display:** Monitor with a minimum resolution of 1366 x 768 pixels (1920 x 1080 Full HD recommended for optimal viewing of large data grids and reports).
- **Network:** Reliable internet or intranet connection to communicate with the application server.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or modern Linux distributions.
- **Web Browser:** An up-to-date modern web browser is required. Recommended browsers include:
    - Google Chrome (latest version)
    - Mozilla Firefox (latest version)
    - Microsoft Edge (latest version)
    - Apple Safari (latest version)
- **Additional Software:** A standard PDF reader (e.g., Adobe Acrobat or built-in browser PDF viewers) to view and print downloaded reports and payslips.

Non-Functional Requirements

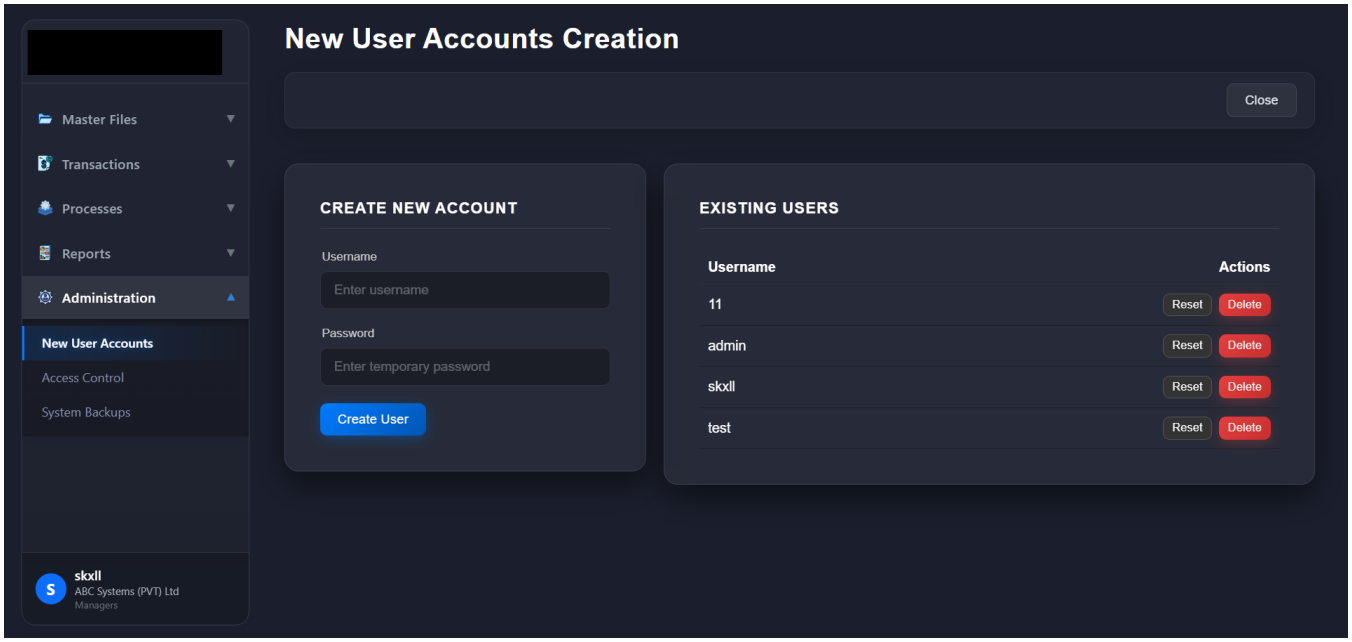
Security

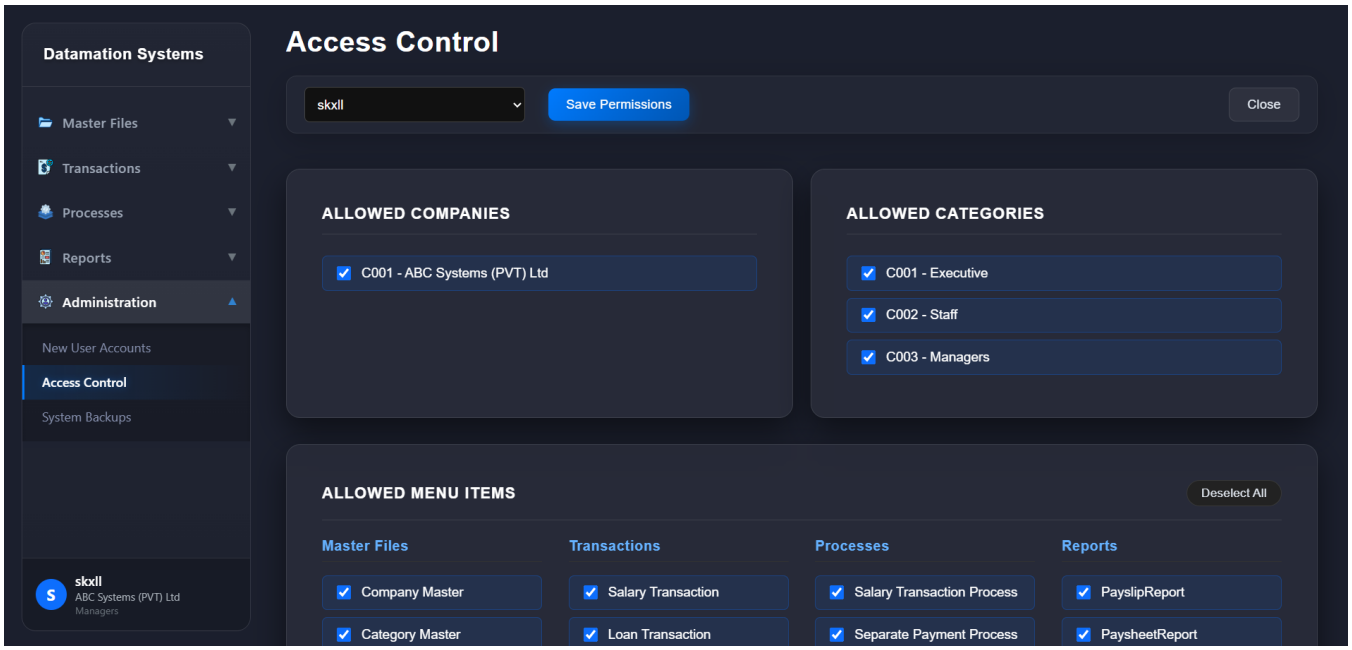
User Creation & Login to the System

- **Account Provisioning:** Only authorized System Administrators will have the capability to create, modify, or deactivate user accounts within the system.
- **Secure Authentication:** The login interface will require users to authenticate using uniquely assigned credentials (e.g., username/email and password). All user passwords must be securely encrypted and hashed before being stored in the database to prevent exposure in the event of a database breach.
- **Session Management:** The application will utilize secure token-based authentication to maintain user sessions. To prevent unauthorized access on unattended terminals, the system will automatically terminate the session and log the user out after a predefined period of inactivity.

Company Access Control

- **Role-Based Access Control (RBAC):** The system will enforce strict RBAC, allowing administrators to configure granular permissions. Users will only have access to the specific menus, screens, and functional operations (e.g., view-only vs. edit/delete capabilities) that are explicitly mapped to their assigned role.
- **Entity-Level Restrictions:** For organizations managing multiple branches, departments, or subsidiary companies, the application will feature dedicated 'Company Access Control'. This ensures that an HR representative or manager assigned to one specific company or department cannot view, edit, or process payroll data for another.
- **Data Segmentation:** Access privileges will be validated at the backend API level, ensuring that even if a user attempts to navigate to a restricted area directly, the system will block the request and protect the underlying data.


New User Accounts Creation
Close
Master Files
Transactions
Processes
Reports
Administration
New User Accounts
Access Control
System Backups
skxll
ABC Systems (PVT) Ltd
Managers
CREATE NEW ACCOUNT
Username
Enter username
Password
Enter temporary password
Create User
EXISTING USERS
Username
Actions
11
Reset
Delete
admin
Reset
Delete
skxll
Reset
Delete
test
Reset
Delete


Datamation Systems
Access Control
skxll
Save Permissions
Close
Master Files
Transactions
Processes
Reports
Administration
New User Accounts
Access Control
System Backups
skxll
ABC Systems (PVT) Ltd
Managers
ALLOWED COMPANIES
C001 - ABC Systems (PVT) Ltd
ALLOWED CATEGORIES
C001 - Executive
C002 - Staff
C003 - Managers
ALLOWED MENU ITEMS
Deselect All
Master Files
Transactions
Processes
Reports
Company Master
Salary Transaction
Salary Transaction Process
PayslipReport
Category Master
Loan Transaction
Separate Payment Process
PaysheetReport

Payroll System Operational Cycle

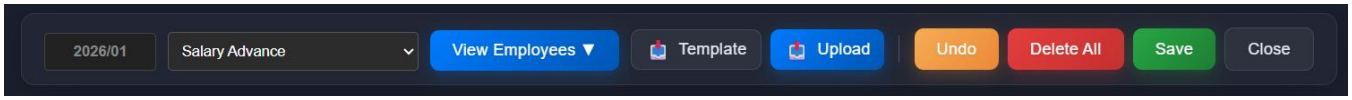
The steps to be followed to run payroll system for a particular month:

1. Check the system date in the Company master file.
2. Add new employees for this month and edit any existing employees' information in the employee master file.
3. Change any master file information if necessary.
4. Do employee promotion and transfers if available.
5. Enter all monthly data to the "salary transaction".
6. Print "Salary Transaction Listing" report & check for data entry errors.
7. Salary Increments
8. Amend or Add records to the loan transaction
9. Process "Salary Transaction Process"
10. Print all the monthly reports.
11. If user finds any data entry errors, correct it and run the month end again.
12. After finalizing the current month salary data print required reports.
13. Do the banks slip transfer process to send employee salary to bank.
14. EPF ETF e Transfer Process
15. After printing all the reports and to get prepare for the next month, run "Month End Process".

User Interface & Controls

Common Tool Bar Operations to maintain a consistent and intuitive user experience, all data entry screens (such as Master Files and Transaction modules) utilize a standardized action toolbar. These common operations dictate how users interact with the underlying database records.

- **Add / New:** Clears the form to create a new database record.
- **Edit:** Unlocks a selected record to allow modifications.
- **Save / Update:** Validates and commits changes securely to the database.
- **Delete:** Removes a record (includes confirmation prompts to protect financial data).
- **Find:** Searches for specific records within the current data grid.
- **Close:** Exits the active module and returns to the previous screen.
- **View Employees:** Displays a list of employee records relevant to the current process.
- **Template:** Downloads a formatted spreadsheet for bulk data entry.
- **Upload:** Imports a completed template for automated mass processing.
- **Delete All:** Clears all staged records in a temporary batch (requires confirmation).
- **Undo:** Reverts the most recent unsaved action or accidental change.



Functionalities

Module: Company Master File, this interface is used to capture and manage the core structural, statutory, and financial settings for a business entity within the payroll system. The data captured here acts as the foundational master data for processing salaries, generating bank files, and filing statutory reports.

General Details Section This section captures the primary identification and location data for the company.

- Company Code (Mandatory): A unique alphanumeric identifier (e.g., "C001") acting as the primary key reference for the company across the entire application.
- Company Name (Mandatory): The full, legally registered name of the business entity (e.g., "ABC Systems (PVT) Ltd").
- Address: The official registered physical address of the company used for correspondence and official documentation.
- Zone Code: An identifier used to categorize the company into a specific operational, geographical, or administrative zone.
- District Code: A numeric or alphanumeric identifier representing the geographical district where the company is located, often required for localized statutory reporting.

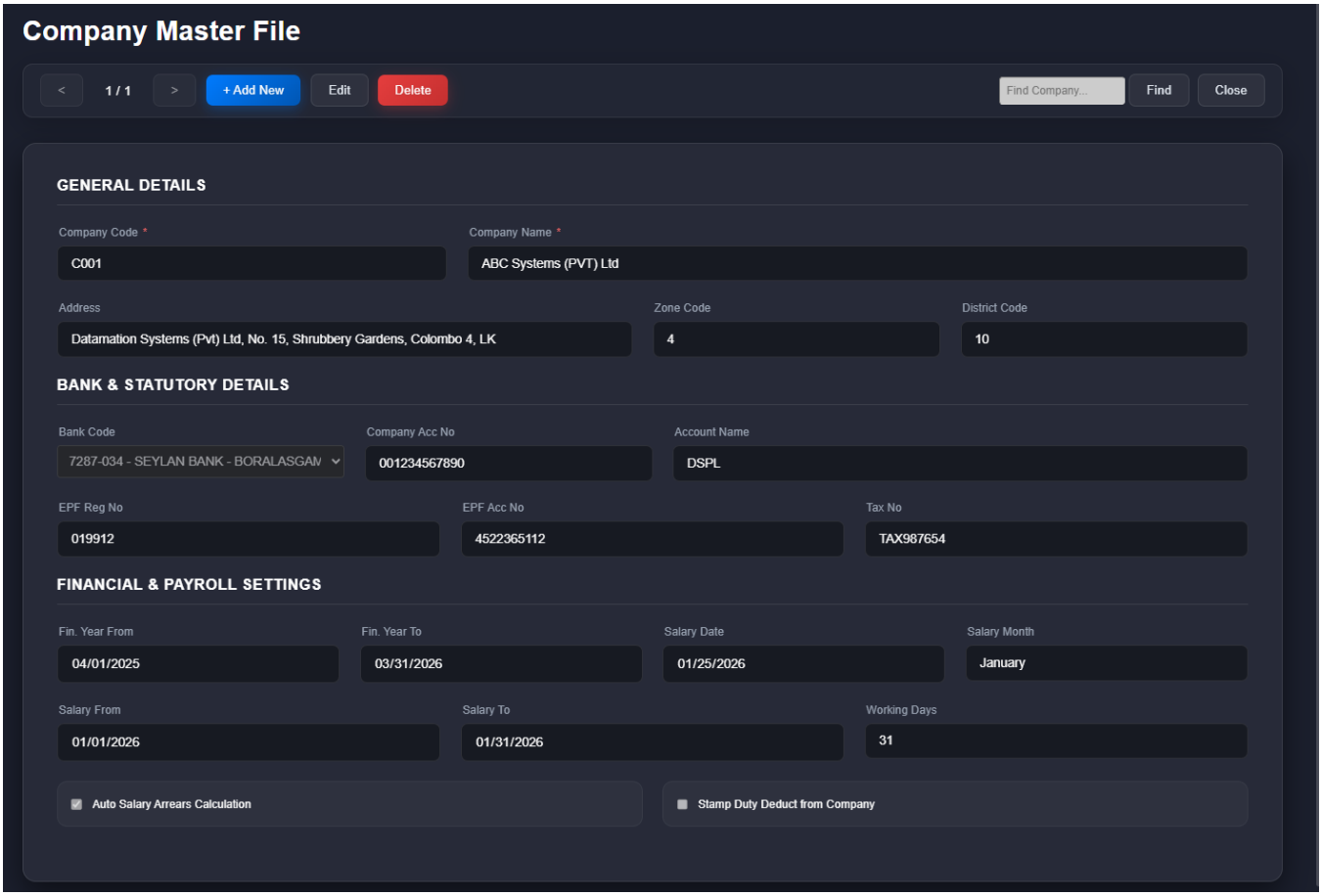
Bank & Statutory Details Section This section stores the financial routing information and government-issued registration numbers necessary for compliance and direct deposits.

- Bank Code: A dropdown or lookup field linking the company to a specific bank and branch (e.g., Seylan Bank) through which payroll fund disbursements will be routed.
- Company Acc No: The official corporate bank account number from which employee salaries and statutory payments will be debited.
- Account Name: The exact legal name registered to the company's bank account.
- EPF Reg No: The Employer Provident Fund registration number assigned to the company by the labor department.
- EPF Acc No: The specific account number associated with the company's EPF contributions and remittance files.
- Tax No: The company's Tax Identification Number (TIN) or VAT number, required for corporate tax and statutory deductions reporting.

Financial & Payroll Settings Section This section defines the active payroll periods, fiscal timelines, and automated calculation rules for the company.

- Fin. Year From: A date field defining the official start date of the company's current financial/fiscal year.

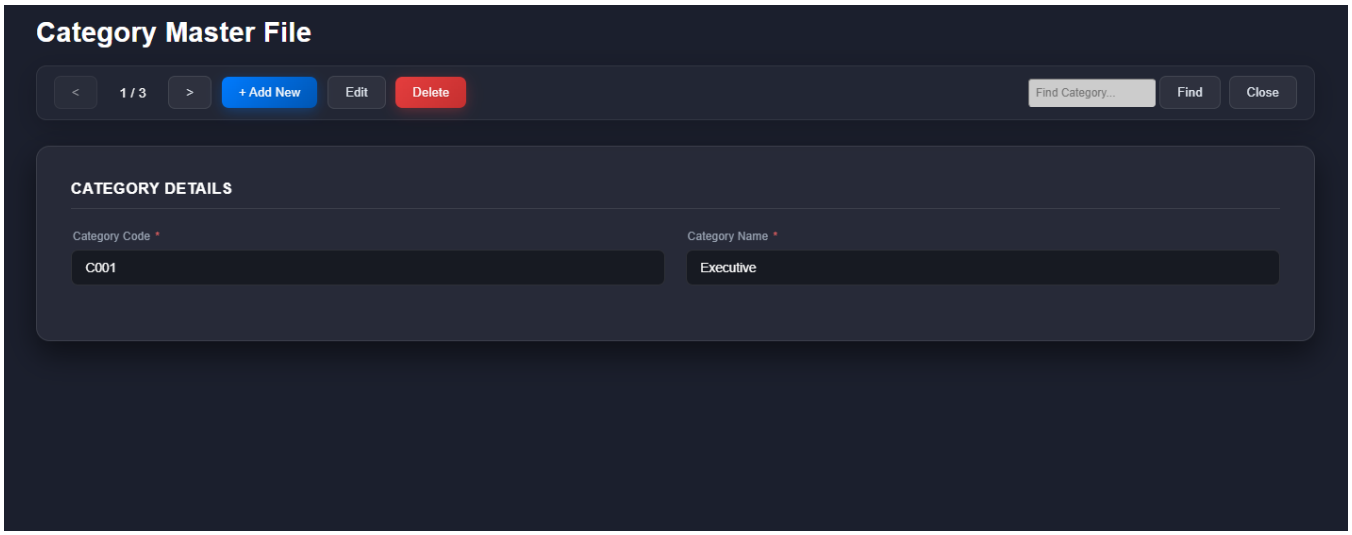
- Fin. Year To: A date field defining the official end date of the company's current financial/fiscal year.
- Salary Date: The standard scheduled date for the physical disbursement of monthly payroll.
- Salary Month: Indicates the current active month that the payroll engine is processing.
- Salary From: The starting date of the current payroll attendance and calculation cycle.
- Salary To: The ending cut-off date of the current payroll attendance and calculation cycle.
- Working Days: The total defined number of billable or payable working days within the current salary period.
- Auto Salary Arrears Calculation (Checkbox): A boolean setting. When enabled, the backend system will automatically calculate and apply any pending back pay, increments, or salary arrears for employees during the monthly payroll execution.
- Stamp Duty Deduct from Company (Checkbox): A boolean setting. When enabled, the statutory stamp duty tax is absorbed as a corporate expense rather than being deducted from the individual employees' net pay.



Module: Category Master File, this interface is used to define and manage employee classifications or groups within the organization (e.g., Executive, Management, Contract, Support Staff). These categories are essential for grouping employees, applying specific payroll rules or benefits, and generating aggregated management reports.

Category Details Section This section captures the core identifying information for the employee classification.

- Category Code (Mandatory): A unique alphanumeric identifier (e.g., "C001") acting as the primary key reference for this specific employee group across the payroll system.
- Category Name (Mandatory): The descriptive, human-readable title of the classification (e.g., "Executive"). This label will appear in employee profiles, dropdown menus, and reporting modules.



Module: Employee Master File, this interface serves as the central repository for individual employee profiles. It aggregates personal identification, organizational placement, compensation parameters, bank routing, and payroll processing rules for each staff member.

Basic Information Section This section captures fundamental personal data and links the employee to the broader organizational structure.

- Company Code (Mandatory): Links the employee to a specific corporate entity defined in the Company Master.
- Category Code (Mandatory): Assigns the employee to a specific classification (e.g., Executive, Contract) defined in the Category Master, determining applicable payroll rules.
- Employee Code (Mandatory): A unique, primary key identifier for the individual employee within the system.
- NIC (Mandatory): The employee's National Identity Card number, essential for legal and statutory identification.
- Initials (Mandatory): The employee's initials or first name(s).
- Surname (Mandatory): The employee's last name or family name.
- EPF No: The employee's specific tracking number for the Employer Provident Fund, used for statutory deductions and reporting.
- Address: The current residential address of the employee.
- Birth Date (Mandatory): The employee's date of birth, used for age validation and retirement calculations.

Employment Details Section This section defines the employee's job role, departmental allocation, and core compensation timeline.

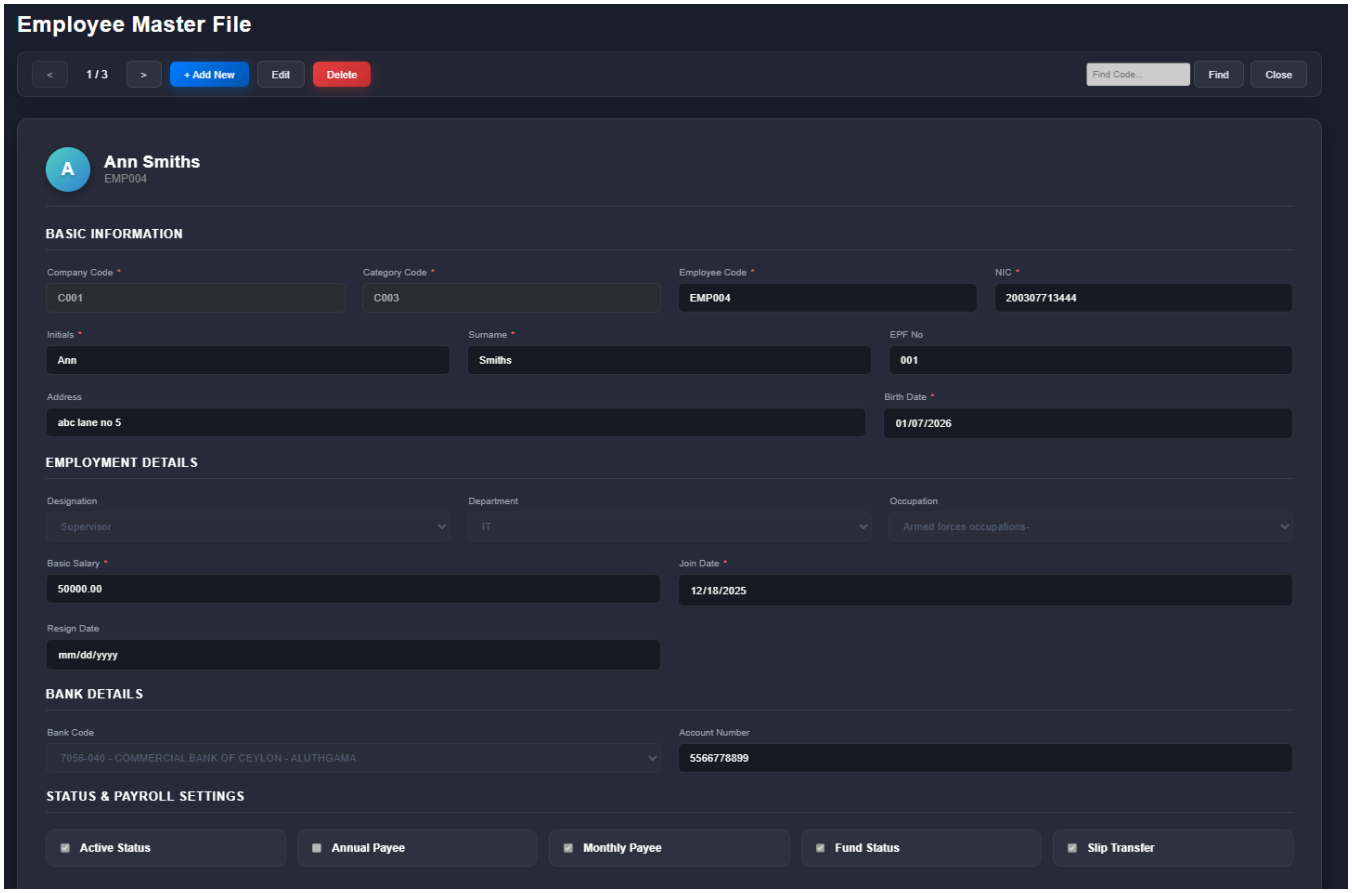
- Designation: A dropdown selecting the employee's official job title (e.g., Supervisor).
- Department: A dropdown assigning the employee to a specific functional group or cost center (e.g., IT).
- Occupation: A dropdown classifying the broader occupational category of the employee, often used for standardized government or statistical reporting.
- Basic Salary (Mandatory): The fixed baseline monetary compensation assigned to the employee before any allowances or deductions are applied.
- Join Date (Mandatory): The official date the employee commenced employment, used to calculate tenure, prorated salaries, and leave entitlements.
- Resign Date: The date the employee's contract was terminated or they resigned. If populated, it acts as a trigger to halt future payroll processing for this individual.

Bank Details Section This section manages the routing information required for automated direct deposits.

- Bank Code: A dropdown linking the employee to a specific bank and branch for salary disbursement.
- Account Number: The employee's personal bank account number where net pay will be credited.

Status & Payroll Settings Section This section contains boolean (checkbox) flags that dictate how the payroll engine handles this specific employee during the processing cycle.

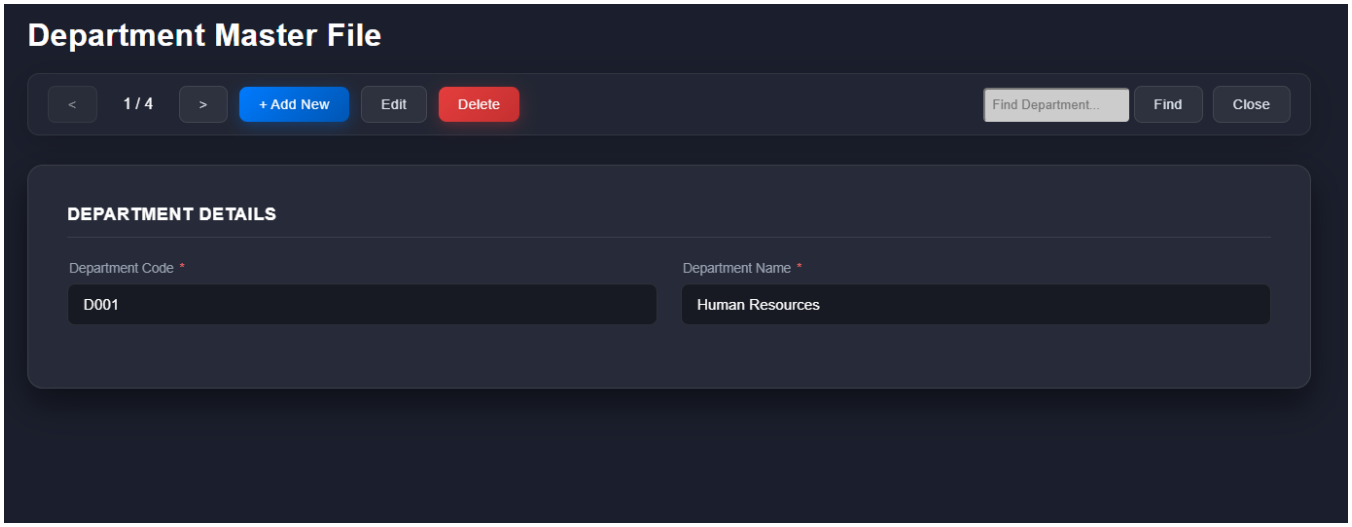
- Active Status: When checked, the employee is considered active and will be included in the current payroll run. If unchecked, they are suspended or inactive and bypassed during processing.
- Annual Payee: Indicates if the employee is subject to annual rather than monthly taxation or salary rules (e.g., specific contractor agreements).
- Monthly Payee: Indicates the employee is on a standard monthly payroll cycle.
- Fund Status: When checked, the system will automatically calculate and deduct statutory fund contributions (like EPF/ETF) for this employee.
- Slip Transfer: When checked, authorizes the automated generation and transfer/emailing of the digital payslip to the employee upon payroll finalization.



Module: Department Master File, this interface is used to define the organizational structure by creating specific departments, divisions, or cost centers within the company. This data allows employees to be grouped logically for management reporting and payroll expense allocation.

Department Details Section

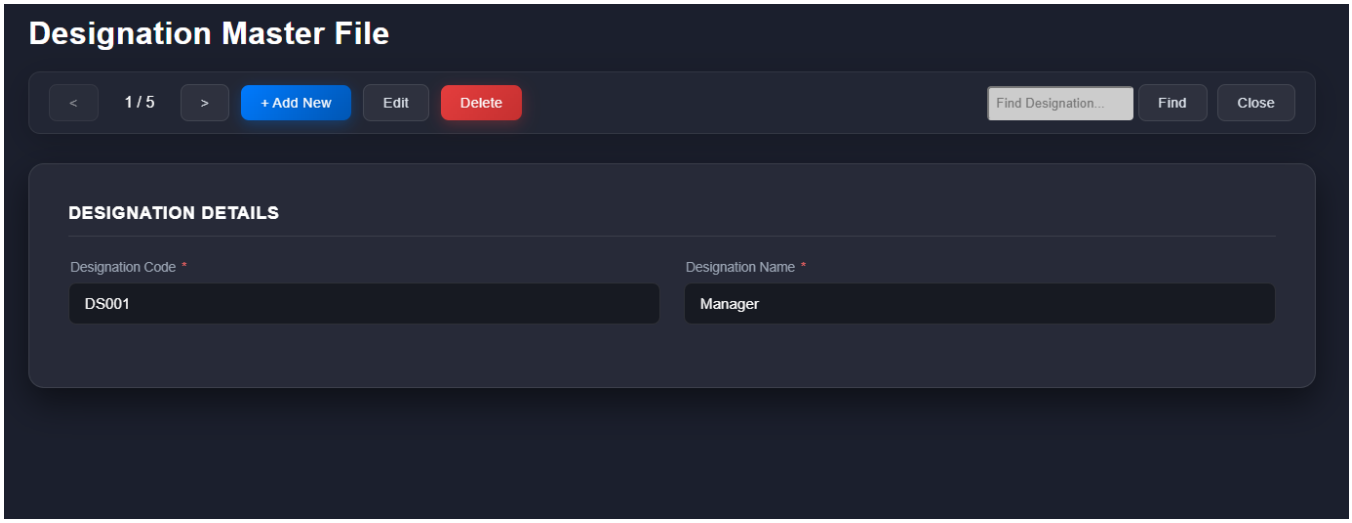
- Department Code (Mandatory): A unique alphanumeric identifier (e.g., "D001") that serves as the primary key for the department in the database.
- Department Name (Mandatory): The standard, human-readable name of the department or cost center (e.g., "Human Resources", "Finance", "IT"). This will populate dropdowns in the Employee Master File and reporting modules.



Module: Designation Master File, this interface manages the standardized list of job titles and organizational roles available within the company.

Designation Details Section

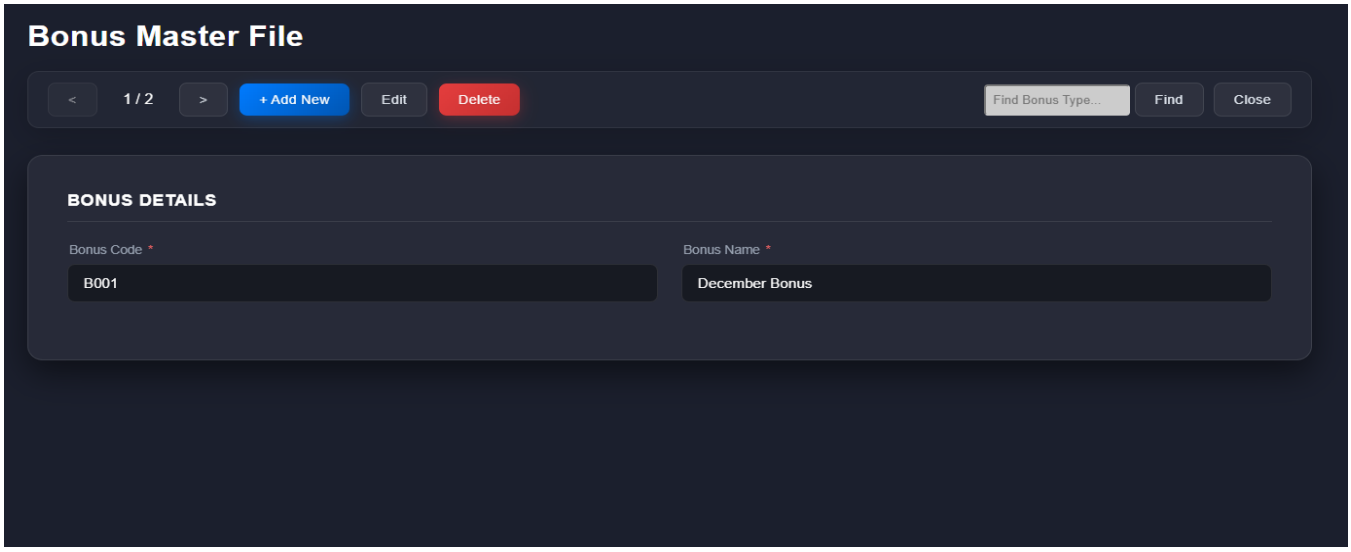
- Designation Code (Mandatory): A unique alphanumeric identifier (e.g., "DS001") acting as the primary key for the specific job role.
- Designation Name (Mandatory): The official job title (e.g., "Manager", "Software Engineer", "Clerk"). This standardized list prevents data entry inconsistencies when assigning roles to employees.



Module: Bonus Master File, this interface defines the various types of non-standard compensation or bonuses that the company can award to employees. Establishing these categories allows the payroll engine to track, calculate, and report on different bonus structures accurately.

Bonus Details Section

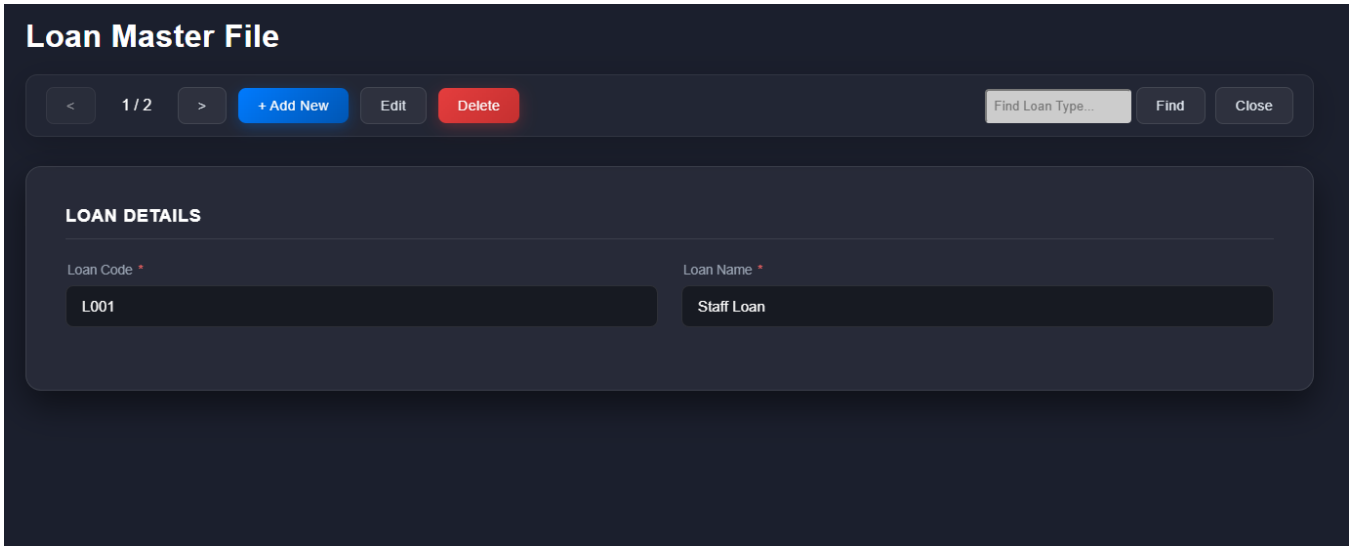
- Bonus Code (Mandatory): A unique alphanumeric identifier (e.g., "B001") used by the backend system to process specific bonus calculations.
- Bonus Name (Mandatory): The descriptive name of the bonus type (e.g., "December Bonus", "Performance Bonus", "Annual Retention Bonus"). This is the label that will appear on the employee's final payslip.



Module: Loan Master File, this interface is used to categorize the different types of financial loans, advances, or credit facilities that the company issues to its employees. This is crucial for managing the distinct recovery rules and deduction schedules associated with different loan types.

Loan Details Section

- Loan Code (Mandatory): A unique alphanumeric identifier (e.g., "L001") representing the specific loan category within the system.
- Loan Name (Mandatory): The descriptive name of the loan type (e.g., "Staff Loan", "Vehicle Loan", "Festival Advance"). This categorizes the deduction when it is processed against the employee's monthly payroll.



Module: Salary Item Master File, this interface is the configuration hub for all financial components that make up an employee's payslip. It allows administrators to define custom earnings, deductions, and allowances, specifying exactly how each item should behave and be calculated during the payroll processing cycle.

Item Details Section

This section defines the core attributes and classification of the pay component.

- Company Code (Mandatory): Links the salary item to a specific company entity (e.g., "C001").
- Type (Mandatory): A dropdown classification (e.g., "Earnings", "Deductions") that dictates whether this item will add to or subtract from the employee's gross pay.
- Item Code (Mandatory): A unique identifier for the specific pay element (e.g., "SI105"). This code is heavily used as a reference variable in complex formula building.
- Item Name (Mandatory): The descriptive, human-readable label (e.g., "OT Amount") that will be printed on the employee's final payslip and management reports.
- Pay Date: An optional date field. Used if this specific salary item is scheduled to be disbursed on a date separate from the standard company salary date.

Formula Configuration Section

This is the central logic gate for automated payroll processing.

- Formula Configuration (Text Box): A read-only or strictly formatted field that displays the underlying logic structure for the salary item. It is stored as a stringified JSON array, which is optimal for relational database storage and allows a JavaScript-based backend engine to easily parse and evaluate the dynamic expressions during the monthly payroll run.
- Configure Button: Triggers the "Formula Builder" modal, providing a user-friendly interface to construct the JSON logic without requiring manual code entry.

Compliance & Behavior Flags

These boolean (checkbox) settings define how the item interacts with statutory bodies and general payroll behavior.

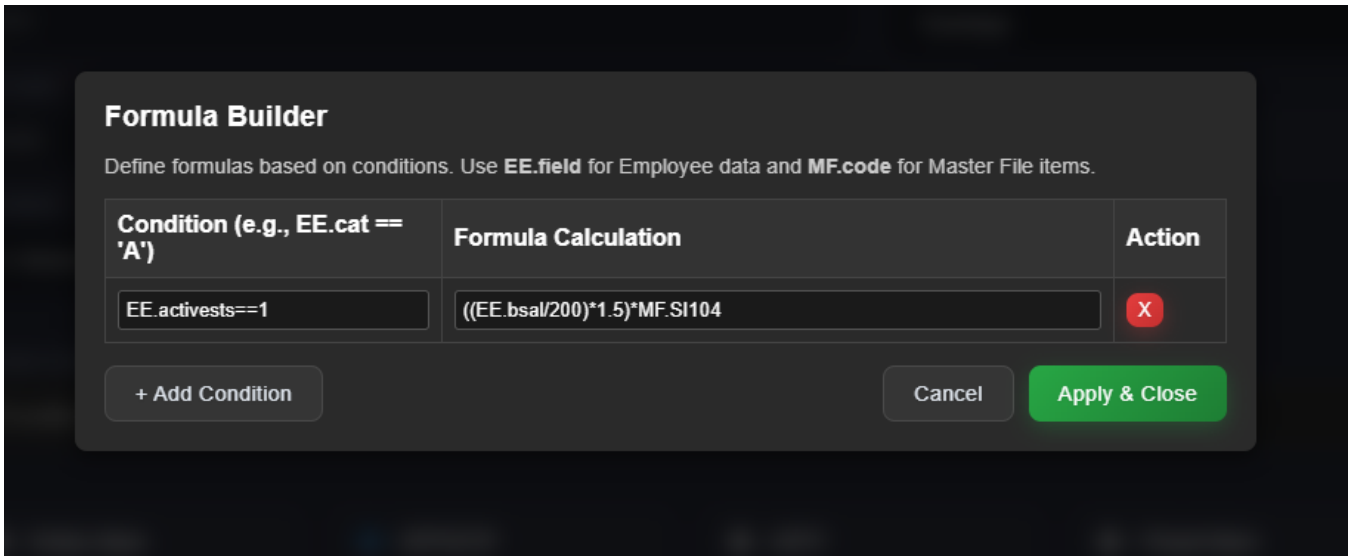
- Entry Item: If checked, this item is not calculated via a formula. Instead, its monetary value is manually entered (or bulk uploaded) each month before payroll processing.
- EPF/ETF: If checked, the monetary value of this item will be included in the gross amount used to calculate statutory Employer Provident Fund and Employees' Trust Fund contributions.
- APIT: If checked, this item is subjected to Advance Personal Income Tax calculations.
- Fixed Item: Indicates that once an amount is set or calculated for an employee, it remains a constant recurring figure in subsequent payroll cycles until manually changed.
- Separate Pay Item: If checked, the system will process and generate a distinct bank transfer file and payslip for this specific item, isolating it from the standard monthly salary.

- Deactivate Item: Acts as a soft-delete mechanism. If checked, the salary item is retained in the database for historical reporting but is ignored by the payroll engine in future calculation runs.

Component: Formula Builder (Modal) A graphical interface accessed via the "Configure" button. It allows administrators to build conditional calculation logic (IF/THEN scenarios) mapping employee data (EE.field) and other master file values (MF.code) to generate dynamic pay amounts.

Formula Definition Grid

- Condition (Input Field): The logical rule that must be met for the formula to execute. It utilizes system variables (e.g., EE.activests==1 checks if the Employee Active Status is true).
- Formula Calculation (Input Field): The mathematical expression executed if the condition is met. It supports standard arithmetic operators and variable fetching.
    - Example Breakdown: ((EE.bsal/200)*1.5)*MF.SI104 fetches the Employee Basic Salary (EE.bsal), divides it by standard working hours (200), multiplies by the overtime rate (1.5), and multiplies the result by the actual OT hours logged for that period (MF.SI104).
- Action (X) Button: Removes the specific condition row from the builder.
- + Add Condition Button: Appends a new row to the grid, allowing administrators to chain multiple scenarios together (e.g., creating tiered commission structures or varying overtime rates based on different conditions).
- Cancel Button: Discards any changes made in the modal and closes it.
- Apply & Close Button: Validates the inputted syntax, serializes the rows into the final JSON array format, populates the parent "Formula Configuration" field, and closes the modal.



Salary Item Master File

< 9 / 9 > + Add New Edit Delete Find Salary Item... Find Close

ITEM DETAILS

Company Code *
C001

Type *
Earnings

Item Code *
SI105

Pay Date
mm/dd/yyyy

Item Name *
OT Amount

Formula Configuration
[{"condition":"EE.activests==1","formula":"((EE.bsal/200)*1.5)*MF.SI104"}]
Configure

- [ ] Entry Item
- [x] EPF/ETF
- [ ] APIT
- [ ] Fixed Item
- [ ] Separate Pay Item
- [ ] Deactivate Item